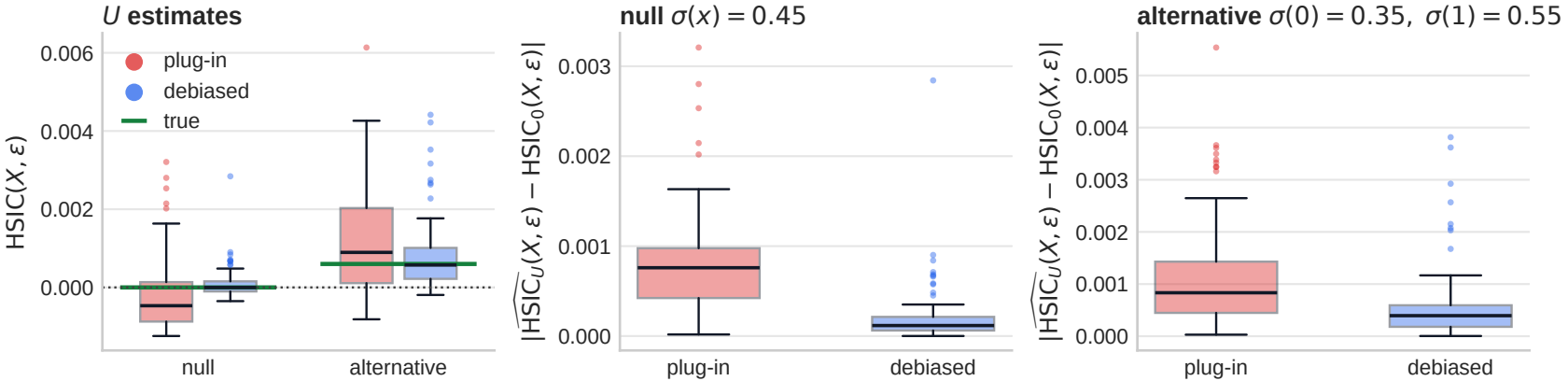
Cross-fit U statistics for $\mathrm{HSIC}(X, \varepsilon)$
U estimates
plug-in
debiased
true
$\mathrm{HSIC}(X, \varepsilon)$
0.006
0.004
0.002
0.000
null
alternative
null $\sigma(x) = 0.45$
$|\widehat{\mathrm{HSIC}}_U(X, \varepsilon) - \mathrm{HSIC}_0(X, \varepsilon)|$
0.003
0.002
0.001
0.000
plug-in
debiased
alternative $\sigma(0) = 0.35,\ \sigma(1) = 0.55$
$|\widehat{\mathrm{HSIC}}_U(X, \varepsilon) - \mathrm{HSIC}_0(X, \varepsilon)|$
0.005
0.004
0.003
0.002
0.001
0.000
plug-in
debiased